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Chapter 13
Poker Made Simple: The AKQ Game

We continue our investigation of game theory in poker games with a game that represents perhaps the simplest form of symmetrical, non-trivial poker. Our first exposure to this game was a Card Player magazine article by Mike Caro. We suppose that he intended the game as a simplified object lesson in betting and bluffing ratios, and in that capacity it serves quite well. We have also expanded on this by studying this beyond the limit case it was first presented in, and in doing so, discovered something surprising about no-limit poker. But for now we concern ourselves with the half-street limit case for this game.

The fundamental elements of the game are these:

- There is a three-card deck, containing an Ace, a King, and a Queen.
- The players are each dealt one card without replacement and there is an initial ante.
- There is a round of betting, after which there is a showdown (if neither player folds). In the showdown, the high card wins.

The three-card deck introduces a topic that we have not so far considered in our games - ***card removal.*** In the [0,1] game both players had symmetric distributions, and one player's card did not affect his opponent's card. However, in this game, if you hold the ace, then the opponent holds either the king or the queen. Likewise, if you hold the queen, the opponent holds the king half the time and the ace half the time.

Example 13.1 - AKQ Game #1
One half street.
Pot size of 2 units.
Limit betting of 1 unit.

The complete ex-showdown payoff matrix (from Y's point of view) for this game as follows:

	Y	**Ace**		**King**		**Queen**	
X		Bet	Check	Bet	Check	Bet	Check
Ace	Call			-1	0	-1	0
	Fold			+2	0	+2	0
King	Call	+1	0			-1	0
	Fold	0	0			+2	0
Queen	Call	+1	0	+1	0		
	Fold	0	0	0	0		

Our goal is to find the optimal strategies for this game. Immediately we can reduce this game by eliminating dominated strategies. Comparing the options when X holds an ace, we can see that calling has expectation greater than or equal to that of folding in all cases. Thus folding an ace is dominated by calling with it. As a result of this, we can eliminate the strategic option of folding an ace from X's strategy options, since it is dominated by calling with an ace. In the same way, we can see that calling with a queen is dominated by folding a queen. So we can eliminate calling with a queen from the strategic options, leaving us with a simplified game:

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	Y	**Ace**		**King**		**Queen**	
X		Bet	Check	Bet	Check	Bet	Check
Ace	Call			-1	0	-1	0
King	Call	+1	0			-1	0
	Fold	0	0			+2	0
Queen	Fold	0	0	0	0		

Now we can turn our attention to strategies that are dominated for Y. We can see that betting a king is dominated by checking it for Y, as well as checking an ace. So we can further simplify our game by eliminating these alternatives:

	Y	**Ace**	**Queen**	
X		**Bet**	**Bet**	**Check**
Ace	Call		-1	0
King	Call	+1	-1	0
	Fold	0	+2	0

Let's take a moment to intuitively discuss what's occurred here. First, we considered X's options. We found two strategies where X could always do at least as well: calling with his aces, and folding his queens to bets. Hopefully this is obvious - we never fold the nuts to a bet on the river, nor call when we could fold with a completely hopeless hand. Since X would never take these actions and can never gain by taking them in the future, we can simply eliminate them from the game matrix.

Next, we took the reduced game, and applied the same reasoning to Y's choices. If Y has a king, he knows he can never bluff X's aces out, and can never be called by X's queens. And if he has an ace, he can never lose by betting, and he might get a call from a king.

By eliminating the dominated strategies, we are left with a much more simplified game - in fact, from a strategic option standpoint, this game is a 2x2 matrix, just as games in Chapter 13 were. Here we see the oscillation of strategies from a pure standpoint, as we described there.

If X calls all the time with kings, Y's best response is never bluffing and only value betting. In turn, X responds to that by folding all kings. Y exploits that by bluffing all the time; and X returns to his original strategy of calling all the time with kings. This indicates that we have an equilibrium between these strategies in mixed strategies.

We can actually create a 2x2 matrix by determining the equities of the undominated strategies against each other:

The two undominated strategic options are whether Y should bluff queens, and whether X should call with Kings.

We End Y's equity from the two options by considering the possible hands that X can hold and the outcomes.

We can name the four strategic options:

Y_1 : Y bluffs with queens
Y_2 : Y checks queens.
X_1 : X calls with kings

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X_2 : X folds kings.

We can find the expectation of each of these matchups:

Υ_1 vs. X_1:

When Y has an ace, he gains a bet against X's kings and nothing against X's queens. When Y has a king, no betting occurs. When Y has a queen, he bluffs and loses a bet to X's aces and kings.

Each card matchup occurs $^1/_6$of the time; Y gains a bet in one, and loses a bet in two for a total of:
$< \Upsilon_1 , X_1 > = - {}^1/_6$

Υ_2 vs. X_2:

When Y has an ace, he gains a bet against X's kings and nothing against X's queens.
When Y has a king, no betting occurs. When Y has a queen, no betting occurs.

Y gains a bet in one case, and does not lose a bet, for a total of:

$< \Upsilon_2 , X_1 > = + {}^1/_6$

Υ_1 vs. X_2:

When Y has an ace, he never gains a bet. (X folds.) When Y has a king, no betting occurs.
When Y has a queen, he bluffs and wins the pot (of two bets) when X has a king, and loses a bet when X has an ace.

Y gains two bets and loses one, for a total of:

$< \Upsilon_1 , X_2 > = + {}^1/_6$

Υ_2 vs. X_2:

When Y has an ace, he never gains a bet (X folds). When Y has a king, no betting occurs.
When Y has a queen, no betting occurs, and so there is no net impact.

$< \Upsilon_2 , X_2 > = 0$

This results in the following *2x2* matrix:

	Y bets queens	**Y checks queens**
X calls with kings	$- {}^1/_6$	$+ {}^1/_6$
X folds kings	$+ {}^1/_6$	0

Multiplying all the cells of a payoff matrix by a positive constant does not change the strategic implications of the game. Multiplying by -1 simply swaps the payouts between the two players.

If we multiply the AKQ Game #1 matrix by -6, we have:

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	Y bets queens	**Y checks queens**
X calls with kings	+1	-1
X folds kings	-1	0

This matrix should be familiar to the reader from Chapter 10; recall that the payoff matrix of Cops and Robbers was:

	Robber robs	**Robber stays home**
Cop patrols	+1	-1
Cop doesn't patrol	-1	0

This game is identical, then, to Cops and Robbers. We know from the analysis of that game that the optimal strategies are $^1/_3$ and $^1/_3$, and we can compare this result to a result we will find directly.

We can also solve the game by finding for each player a strategy that makes his opponent indifferent between his two strategic options. Consider Y, who must decide whether to bluff with a queen. His ex-showdown EV from the two actions he can take with a queen (using *c* as X's calling frequency with a king) is:

<Y, bluff > = (lose to aces) + (lose to kings) -f- (kings fold, win the pot)
<Y, bluff > = *p*(X has an ace) (lose one bet) + *p*(X has a king) *p*(X calls) (lose one bet) +
p(X has a king) *p*(X folds) (the pot)
<Y, bluff > = ($^1/_2$)(-1) + ($^1/_2$)(*c*)(-1) + ($^1/_2$)(1 – *c*) (2)
<Y, bluff > = $^1/_2$ (-1) + $^1/_2$ [*c* (-1) + (1 - *c*) (2)]
<Y, bluff > = - $^1/_2$ - $^1/_2$ *c* + 1 - *c*
<Y, bluff > = $^1/_2$ - $^3/_2$ *c*
<Y, check> = 0

Setting these two things equal:

$^1/_2$ - $^3/_2$ *c = 0*
c = $^1/_3$

Next, set X to be indifferent to calling with a king. Using *b* as Y's bluffing frequency with a queen:

<X, call > = (lose to aces) + (win against queens)
<X, call > = $^1/_2$ (-l) + $^1/_2$ [*b*(1)]
<X, call > = - $^1/_2$ + $^1/_2$ *b*
<X, fold > = $^1/_2$(*b*) (-2)
<X, fold > = - *b*
- $^1/_2$ + $^1/_2$*b* = - *b*
b = $^1/_3$

So we have the optimal strategy pair: X should call $^1/_3$ of the time when he has kings, and Y should bluff $^1/_3$ of the time when he has queens.

Let's again take a moment to discuss intuitively what's going on here. The idea for Y is to get value from his set of hands. In an exploitive sense, if he knew what strategy X were going to

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employ with his kings, he could bet or bluff accordingly. However, let's say that he doesn't and he wants to simply get all his value no matter what the opponent does.

There are two sources of value for him: stealing pots with queens, and getting paid off with aces. So he wants to create a situation where the value he gets from each of these options is equal. Then his opponent won't be able to exploit him. What he does, then, is bluff with a frequency so that the value of his whole strategy is the same no matter what the opponent does; specifically, no matter whether the opponent calls with a king or not. It turns out for this situation that the proper ratio of aces to bluffed queens is three to one; hence he should bet $^1/_3$ of his queens, since he will bet all his aces. Recall that in Chapter 11, we called this ratio of bluffs to bets α, and showed that it had the value $1/(P+1)$. (See Equation 11.1) For the pot of two bets, we can see that this ratio indeed calculates to $^1/_3$.

Similar reasoning works for X. He wants to call often enough that the opponent is indifferent to bluffing with his queens. When his opponent bluffs successfully, he wins two bets, while bluffing unsuccessfully costs just one bet. So X must call twice as often as he folds in order to make the bluffer indifferent. This fraction is $^2/_3$ (leaving $^1/_3$ of hands to fold). But half of the time that the opponent is bluffing, X will hold an ace. So he needs to call with an additional fraction of his hands equal to $^2/_3 - {}^1/_2$, or $^1/_6$. Since kings make up $^1/_2$ of his hands when the opponent has a queen, he needs to call with $^2/_6$ or $^1/_3$ of kings. Again recall in Chapter 11, we showed that X would fold α of the time with hands that could beat a bluff. Here again, he folds $^1/_3$ of the total hands he holds that can beat a bluff.

Y gains value in this game, as you would expect; after all, he could just check behind all the time to get zero value, and the betting and bluffing is valuable to him. We can calculate the value of the game to Y by evaluating the strategies and outcomes. We could calculate all the fractional possibilities for each action, but instead we can use a shortcut. Recall that when the players play optimal strategies, X is indifferent to calling; that is, he does equally well calling or folding. So we can simply pretend that he folds all the time. Now if he actually did fold all the time, he would be exploitable. But since we're only calculating the value of the game and not the strategies, we can simply hold Y's strategy constant.

The value of the game to Y is actually $^1/_{18}$ of a bet. We invite the reader to verify that this is the value of the game for all strategies that X may employ as far as calling with kings.

We have now been able to solve this half street game, finding the optimal strategies, for a single pot size (2 bets) - however, we can use the same methodology as we employed to solve it for any arbitrary pot size. If we restate the game with an arbitrary pot size P, we have the easily solvable AKQ Game #2.

Example 13.2 - AKQ Game #2

One half street.
Pot size of P units.
Limit betting of 1 unit

Again, we have the complete game payoff matrix:

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Y		Ace		King		Queen	
X		Bet	Check	Bet	Check	Bet	Check
Ace	Call			-1	0	-1	0
	Fold			+P	0	+P	0
King	Call	+1	0			-1	0
	Fold	0	0			+P	0
Queen	Call	+1	0	+1	0		
	Fold	0	0	0	0		

This matrix can be reduced by eliminating dominated strategies, just as in AKQ Game #1 (Example 13.1), yielding:

	Y	Ace	Queen	
X		Bet	Bet	Check
Ace	Call		-1	0
King	Call	+1	-1	0
	Fold	0	+P	0

Our two indifference equations are modified as follows (assume b is Y's bluffing frequency with queens, and c is X's calling frequency with Icings):

<Y, bluff > = (lose to aces) + (lose to kings) + (kings fold, win the pot)
<Y, bluff > = $(\frac{1}{2})(-1) + (\frac{1}{2})(c)(-1) + (\frac{1}{2})(1 - c)(P)$
<Y, bluff > = $\frac{1}{2}(-1) + \frac{1}{2}[c(-1) + (1 - c)(P)]$
<Y, bluff > = $-\frac{1}{2} - \frac{1}{2}c + \frac{1}{2}P - \frac{1}{2}Pc$
<Y, check > = 0

Setting these two things equal:

$-\frac{1}{2} - \frac{1}{2}c + \frac{1}{2}P - \frac{1}{2}Pc = 0$
$1 + c - P + Pc = 0$
$(P + 1)c = P - 1$
$c = (P - 1)/(P + 1)$

This value of *c* is X's calling frequency with kings. When Y has a queen, X holds half aces and half kings. So the total fraction of his hands (aces and kings) with which he calls is $\frac{1}{2} + (\frac{1}{2})(P - 1)/(P + 1)$. This in turn means that he folds $1/(P + 1)$, or the now-familiar constant α, of his hands, to keep Y indifferent from bluffing.

In a similar manner, we have:

<X, call > = (lose to aces) + (win against queens)
<X, call > = $\frac{1}{2}(-1) + \frac{1}{2}[b(1)]$
<X, call > = $-\frac{1}{2} + \frac{1}{2}b$
<X, fold > = $\frac{1}{2}(b)(-P)$

Again, setting these two equal:

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$-\frac{1}{2} + \frac{1}{2}b = \frac{1}{2}(b)(-P)$
$-1 + b = -bP$
$b(1+P) = 1$
$b = 1/(1+P)$
$b = \alpha$

And again, we see that the bluffing to betting ratio for Y here should be α as well.

We can also quickly calculate the value of the game to Y:

Hand Matchup	**Value to second player**	***p*(matchup)**	**Weighted Value**
(A, K)	0	$\frac{1}{6}$	0
(A, Q)	-1/(P+ 1)	$\frac{1}{6}$	$-\frac{1}{6}(P+1)$
(K, A)	0	$\frac{1}{6}$	0
(K, Q)	$+P/(P+1)$	$\frac{1}{6}$	$P/6(P+1)$
(Q, A)	0	$\frac{1}{6}$	0
(Q, K)	0	$\frac{1}{6}$	0
Total			**$(P-1)/6(P+1)$**

So the overall value of the game to Y is $(P-1)/6(P+1)$. We can see that as the pot size goes to infinity, the value of this game converges on $\frac{1}{6}$. If the pot is extremely large, then X will simply call with just about all his kings, and Y will almost never bluff, making the case where Y holds an ace and X a king worth a full bet to Y. This case occurs $\frac{1}{6}$ of the time.

One additional comment must be made - the above solution is only valid if the pot size is greater than 1. The reason is that when the pot is less than 1. Y loses money by bluffing queens - X calls with aces, which make up half his hands. Since Y gains the pot when X folds a king, and loses a full bet when X calls with an ace, bluffing always loses money, even when X folds all his kings. The entire game changes at that point. As the pot approaches one from the positive side, the bluffing ratio rises - at 2, it's $\frac{1}{3}$, at 1.5, it's $\frac{2}{5}$, at 1.1 it's $\frac{10}{11}$, and then at precisely 1, Y stops bluffing.

We can also use a graph to see the value of the game as the size of the pot changes. Notice that the graph suddenly changes at $P = 1$; Y goes from never bluffing to bluffing half of his hands. This happens because calling suddenly becomes a mixed strategy, as does bluffing. Below this pot size, neither side bluffs nor calls; this cannot be exploited because of the frequency of aces, in the opponent's distribution. We can also see at large pot sizes that X has to call frequently while Y can bluff infrequently, as a result, the game favors Y as the pot size increases.

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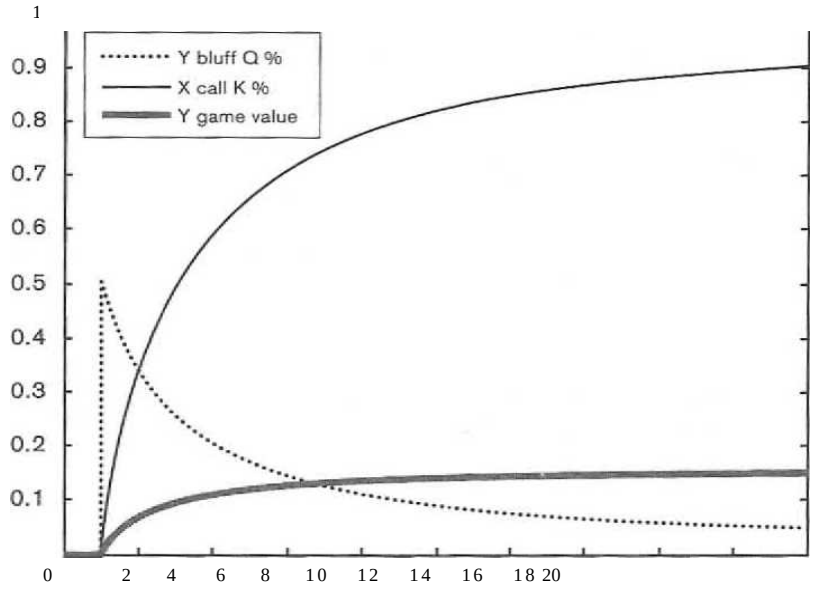

Figure 13.1. Optimal strategies for the limit AKQ game by pot size.

The important lessons here are that changing the size of the pot impacts the players' strategies and their equities; also solving games with specific pot sizes is often just as easy as solving them for arbitrary pot sizes. In the future, we will generally forego solving games with specific pot sizes unless it is necessary for the discussion, preferring to simply treat games as having pot sizes of *P*.

Key Concepts

- Betting and bluffing ratios are the key to optimal play when both sides have clearly defined hand values (i.e., they have the nuts, a hand in the middle, or a dead hand).
- In limit poker, it is virtually always wrong to bet hands in second position on the river that can beat only a bluff but will be called by only better hands.
- Card removal can have a significant impact on the shape of the opponent's distribution.

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